Lab 7: Fuzz Testing with Hypothesis

Student SRN: PES2UG24CS013
Repository: L6_L7_11_amusement_park_ride_access
Objective: Discover edge-case input failures in three ride-access functions, harden validation, and verify the changes with property-based tests.

Functions tested

validate_height_and_fastpass parses a height and fastpass choice; compute_dispatch_throughput calculates riders per minute; resolve_safety_restraint_code maps a zone prefix to a restraint type.

Before the fix

The starter functions raised exceptions for malformed strings, missing separators, invalid numeric values, unknown prefixes, and None. Hypothesis reduced failures to compact examples including ":", "/", and None.

Before fix: processor.py sample run

```
==== Running validate_height_and_fastpass ====
Enter rider spec (Height:FastPass, e.g. 140:YES): Height 140cm: Approved (FastPass Priority)

==== Running compute_dispatch_throughput ====
Enter riders/mins (e.g. 48/2.5): Throughput (riders/min): 19.2

==== Running resolve_safety_restraint_code ====
Enter ride zone code (e.g. CST_01, WTR_A): Restraint: Over-the-Shoulder Harness
```

Before fix: Hypothesis finds input crashes

```
FFF                                                                      [100%]
================================== FAILURES ===================================
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\aarun\AppData\Local\Temp\lab7_before_12h441lg\test_processor.py", line 5, in test_validate_height_and
  |     def test_validate_height_and_fastpass_no_crash(s):
  |                   ^^^
  |   File "C:\Users\aarun\AppData\Local\Programs\Python\Python312\Lib\site-packages\hypothesis\core.py", line 2334, in wr
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\aarun\AppData\Local\Temp\lab7_before_12h441lg\test_processor.py", line 7, in test_validate_height_a
    |     validate_height_and_fastpass(s)
    |   File "C:\Users\aarun\AppData\Local\Temp\lab7_before_12h441lg\processor.py", line 3, in validate_height_and_fastpas
    |     height = int(height_str)
    |              ^^^^^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
...
```

Hardening and verification

The functions now validate input type and structure before parsing. Invalid rider specifications return a clear message, invalid throughput metrics return None, and unknown zones return “Unknown ride zone”. Valid input behavior is retained. Throughput rejects negative/non-finite rider counts, zero/negative/non-finite cycle durations, and overflowed results.

The Hypothesis tests exercise arbitrary strings and None for all three functions. The starter produced three failing fuzz tests; after the fixes, all three tests pass.

After fix: processor.py sample run

```
==== Running validate_height_and_fastpass ====
Enter rider spec (Height:FastPass, e.g. 140:YES): Height 140cm: Approved (FastPass Priority)

==== Running compute_dispatch_throughput ====
Enter riders/mins (e.g. 48/2.5): Throughput (riders/min): 19.2

==== Running resolve_safety_restraint_code ====
Enter ride zone code (e.g. CST_01, WTR_A): Restraint: Over-the-Shoulder Harness
```

After fix: Hypothesis suite passes

```
...                                                                      [100%]
3 passed in 1.44s
```

Reproduction

```
python processor.py
python -m pytest -q test_processor.py
```

Implementation files: processor.py and test_processor.py in the Lab 7 folder.